

BX Player

Product Requirements Document (PRD)

BX Player — Product & Engineering Documentation

Document 1 of 6 · Phase 1: Research, Requirements & Planning

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1 Revision History

Version	Date	Author	Description
0.1	2026-07-28	Product & Engineering	Initial draft of Document 1 — Product Requirements Document

Document Status: Draft — Phase 1 (Requirements & Planning) **Applies to Documents:** This PRD is Document 1 of 6 in the BX Player master documentation set (PRD → UI/UX Spec → System Architecture → Database Design → API Documentation → Development Guide). Per the agreed workflow, frontend design (Document 2) does not begin until this PRD is approved.

2 1. Executive Summary

BX Player is a commercial, browser-installable IPTV media player platform delivered as a Progressive Web App (PWA). It allows individual end users to consume IPTV content from playlists and panels they already own (M3U/M3U8, Xstream Codes “XC”, or provider-issued playlist codes), while giving IPTV resellers and platform administrators the tools to provision, monitor, and manage the accounts that use the app.

Unlike a simple playlist-loading media player, BX Player is architected as a three-tier platform:

- **End User** — installs the PWA, registers, links one or more playlists/panels, and streams Live TV, Movies, and Series through a modern, TiviMate/CR-Player-inspired interface.
- **Reseller** — a mid-tier operator who onboards and manages a group of end users, assigning and revoking playlists/subscriptions on their behalf.
- **Super Admin** — the platform owner, with full visibility and control over all resellers, end users, playlists, devices, and platform-wide analytics.

For the initial 6–7 months, BX Player will operate as a **free platform** (no in-app billing) to build a user base, after which paid plans will be introduced. This PRD defines the requirements for that initial free MVP while keeping the data model and role structure ready for monetization.

This is Document 1 of the BX Player documentation set and covers **Phase 1 only**: requirements, scope, personas, and workflow definition. No UI design decisions (Document 2), frontend architecture, backend, or database design are finalized here — those follow in strict sequence per the project’s development strategy.

3 2. Product Vision

To give independent IPTV resellers and their customers a single, professional, installable media player — free of native app-store friction — that feels as polished as TiviMate, IPTV Smarters Pro, and CR Player, while giving operators the account and playlist management tools they need to run a small IPTV business.

BX Player’s differentiation is not the streaming protocol (it does not host or provide content) but the **experience and operations layer** around a user’s existing IPTV subscription:

- Installable in seconds from a browser — no app store approval, no APK sideloading friction.
- One player, three roles — the same codebase serves the end user, the reseller who manages them, and the super admin who oversees the platform.
- Multiple playlist input methods (M3U/M3U8 URL, Xstream Codes credentials, or a provider-issued playlist code) reflecting how IPTV panels are actually sold and shared in this market.
- Optional PIN-based protection on individual playlists, so a shared device can host multiple protected playlists without exposing all of them to every household member.

4 3. Business Objectives

#	Objective	Success Signal
B1	Acquire an initial free user base to validate product-market fit	500–5,000 registered devices within the free period
B2	Prove the reseller model as a distribution channel	Active resellers each managing multiple end users
B3	Establish a stable, professional player experience competitive with TiviMate/IPTV Smarters	Low buffer/error rates; positive qualitative feedback
B4	Prepare the platform for monetization without a rebuild	Subscription/plan fields present in schema from day one, billing logic deferred
B5	Keep infrastructure cost near-zero during validation	Runs on free-tier hosting (Vercel + Supabase/Neon) at MVP scale

Monetization timeline (confirmed): BX Player will be **free for approximately 6–7 months** from launch. No in-app purchases, billing, or paywalls exist in this MVP. The subscription/plan concept is modeled in the data layer (Document 4) so that paid plans can be introduced later without schema migration surprises, but no billing UI or payment gateway is in scope for this phase.

5 4. Product Scope

5.1 4.1 In Scope (MVP — Phase 1–7 as defined in the development strategy)

- PWA installable web application (installable via browser; add-to-home-screen).
- First-run onboarding and registration (Name, Country, Occupation; auto-generated Device ID; issued User ID).
- Three playlist ingestion methods: M3U/M3U8 URL, Xstream Codes (XC), and Playlist Code.

- Optional PIN protection per playlist (4–8 digit PIN + confirmation).
- Full CRUD playlist management with metadata table (name, type, protected flag, date added, status, actions).
- Media player supporting Live TV, Movies, Series, with modern playback controls (subtitles, audio track, speed, aspect ratio, quality, resume).
- Favorites and Continue Watching.
- Global search across channels/movies/series.
- Account/profile screen showing Device ID and User ID (always visible, per requirement).
- Reseller Portal: manage own sub-users, assign/revoke playlists, view sub-user status.
- Super Admin Portal: manage all resellers and end users, platform-wide analytics, device/session oversight.
- Notification center (playlist updated, EPG updated, login failures, etc.) — in-app only for MVP.
- Dark-first, responsive UI across mobile, tablet, and desktop browsers.

5.2 4.2 Explicitly Out of Scope (MVP)

- Native Android/TV/iOS app store builds (PWA only, per confirmed decision; may be revisited later).
- In-app billing, payment gateways, or paid subscription enforcement.
- Content hosting, transcoding, or DRM — BX Player is a **client and management layer only**; all video content is delivered from the user’s own third-party IPTV panel/provider.
- Downloads/offline video playback (listed as optional/future in the original concept; deferred).
- Multi-language localization beyond English (structure allowed for, not delivered).

5.3 4.3 Product Boundary & Responsibility Note

BX Player does not provide, sell, host, or endorse any IPTV channel content. It is a playback and account-management client for playlists and panel credentials that the user or reseller already possesses from their own third-party IPTV provider. This boundary should be reflected in onboarding copy, Terms of Service, and the admin panel (Document 3+), and is treated as a fixed constraint throughout all subsequent documents.

6 5. Stakeholders

Role	Description	Primary Interest
Platform Owner (Super Admin)	Owns and operates the BX Player platform	Overall platform health, growth, and control over all accounts
Reseller	Small-scale IPTV operator who onboards their own customers into BX Player	Ability to manage their customer base without needing Super Admin access
End User	Individual consumer of IPTV content	Reliable playback, easy setup, protected access to paid playlists
Product/Engineering Team	Builds and maintains BX Player	Clear, sequential, unambiguous specifications (this document set)

7 6. User Personas

7.1 6.1 End User — “Umar, the Subscriber”

- Has purchased IPTV access from a reseller or provider outside of BX Player.
- Wants to install BX Player quickly, paste in a URL/credentials/code, and start watching.
- Not technical; expects Netflix/TiviMate-level polish, large poster art, and simple navigation.
- May share the device with family members, so wants sensitive playlists PIN-protected.

7.2 6.2 Reseller — “Bilal, the Small IPTV Operator”

- Resells IPTV panel access to a personal customer base (tens to low hundreds of users).
- Needs to onboard those customers into BX Player, assign each of them a playlist, and disable access if a customer stops paying (offline, outside BX Player, during the free period).
- Does **not** set pricing or plans inside BX Player (confirmed: Reseller manages only their own sub-users’ playlists/subscriptions; monetization/plan-setting is a platform-level, Super-Admin-owned concern once introduced).
- Wants a simple dashboard: who’s active, who’s expired, quick assign/revoke.

7.3 6.3 Super Admin — “The Platform Owner”

- Operates the entire BX Player platform.
- Needs visibility into all resellers and all end users, regardless of which reseller onboarded them.
- Needs to disable any account, inspect device/session activity, and view platform-wide analytics (DAU, watch time, popular channels, buffering errors, device types).
- Will own the future monetization layer (plans/pricing) when billing is introduced after the free period.

8 7. Functional Requirements

Requirements are grouped by module. Each will be elaborated into full UI specifications in Document 2 and API contracts in Document 5; this section defines *what* must exist, not *how* it is built.

8.1 7.1 Onboarding & Registration (End User)

ID	Requirement
FR-1.1	On first install/launch, the app shows a welcome screen before any login/registration form.
FR-1.2	Registration collects Name, Country, Occupation (required); Phone and Email are optional.
FR-1.3	The system auto-generates a unique Device ID (UUID-based, e.g. IPTV-9AF28F18A) on first run and persists it locally.
FR-1.4	The system issues or accepts a User ID, which the user supplies to their Reseller/Provider when subscribing, and which is later used to link the account to a panel/admin record.
FR-1.5	Device ID and User ID must remain permanently visible in a fixed area of the sidebar/account screen, and must be easily copyable.
FR-1.6	After registration, the user is prompted to enter IPTV details (Panel URL/credentials) or import a playlist directly.

8.2 7.2 Playlist Management (End User / Reseller-assigned)

ID	Requirement
FR-2.1	A “Manage Playlists” section provides three horizontally-tabbed sub-sections: M3U Playlist , XC Playlist , Playlist Code .
FR-2.2	M3U tab fields: Playlist Name (required), Playlist URL — .m3u/.m3u8 (required, URL-validated), XMLTV EPG URL (optional), Protect Playlist checkbox.
FR-2.3	XC tab fields: Playlist Name, Username, Password, Host, XMLTV EPG URL (optional), Protect Playlist checkbox.
FR-2.4	Playlist Code tab fields: Playlist Name, Playlist Code, Protect Playlist checkbox.
FR-2.5	When “Protect Playlist” is checked on any tab, two additional fields appear: PIN and Confirm PIN (4–8 digits, must match).

ID	Requirement
FR-2.6	Saving a playlist stores it with metadata (name, type, protected flag, created date, status) and immediately reflects it in that tab's data table.
FR-2.7	Each tab's data table includes columns for Name, Type-specific identifying fields (Host/Username for XC, etc.), Protected indicator, Date Added, Status, and an Actions column (Edit, Delete; Duplicate/Refresh/View reserved for later iteration per the original concept).
FR-2.8	Validation: required-field checks, URL format validation (M3U tab), duplicate playlist name detection, PIN length/format validation, PIN/Confirm PIN match validation.
FR-2.9	Editing a protected playlist requires re-entering its PIN before fields become editable.
FR-2.10	Deleting a playlist requires a confirmation step.

8.3 7.3 Media Player

ID	Requirement
FR-3.1	Content is browsable by category: Live TV, Movies, Series, (Sports/Kids/News as sub-categories of Live TV where EPG/category data supports it).
FR-3.2	Poster-grid browsing with title, duration, rating, and resolution/quality badge (HD/FHD/4K) per item.
FR-3.3	Player controls include: play/pause, ± 10 second seek, subtitle track selection, audio track selection, playback speed, aspect ratio, quality selection, favorite toggle, "add to playlist," info panel, and EPG (for Live TV).
FR-3.4	Player controls auto-hide during playback and reappear on interaction.
FR-3.5	Playback position is saved to support Continue Watching / resume.
FR-3.6	PIN-protected playlists require PIN entry before their content is browsable or playable.

8.4 7.4 Favorites, Continue Watching, Search

ID	Requirement
FR-4.1	Users can favorite/unfavorite any channel, movie, or series episode.

ID	Requirement
FR-4.2	A “Continue Watching” row/section surfaces in-progress content with resume position.
FR-4.3	Global search filters across Movies, Channels, Series, Actors, and Categories with instant/incremental results.

8.5 7.5 Account & Settings (End User)

ID	Requirement
FR-5.1	Account screen displays Name, Country, Occupation, Device ID, User ID, current linked panel, and (once monetization exists) subscription status.
FR-5.2	Settings include Theme (Dark/Light), Language (structure only for MVP), Buffer Size, Auto Play, Hardware Decode toggle, Subtitle Size, Parental Control, and Cache/Storage management.
FR-5.3	Logout clears the active session but preserves the locally generated Device ID.

8.6 7.6 Reseller Portal

ID	Requirement
FR-6.1	A Reseller can view a list of only the End Users they have onboarded (never another reseller’s users, never platform-wide data).
FR-6.2	A Reseller can assign an M3U, XC, or Code playlist to one of their End Users, and revoke/disable it.
FR-6.3	A Reseller can view each of their End Users’ status (Active/Expired/Disabled), Device ID, User ID, and last login.
FR-6.4	A Reseller cannot set platform pricing/plans or affect billing — this remains a Super Admin capability once introduced (confirmed constraint).

8.7 7.7 Super Admin Portal

ID	Requirement
FR-7.1	Super Admin can view/manage all Resellers and all End Users platform-wide.

ID	Requirement
FR-7.2	Super Admin can view per-user details: Name, Country, Occupation, Device ID, User ID, current panel URL, status, subscription, registration date.
FR-7.3	Super Admin can assign/revoke/disable any playlist for any user, and disable any account (Reseller or End User).
FR-7.4	Super Admin has access to platform analytics: Daily Active Users, Watch Time, Popular Channels, Buffer Errors, Device Types, Online Users, Installed PWAs, Active/Expired Panels.

8.8 7.8 Notifications

ID	Requirement
FR-8.1	In-app notification center surfaces: Playlist Updated, New Movies, Subscription Ending (post-monetization), Failed Login, EPG Updated.

9 8. Non-Functional Requirements

Category	Requirement
Installability	Must pass PWA installability criteria (manifest, service worker, HTTPS) on Chrome/Edge/Android and be addable to home screen on iOS Safari.
Performance	Poster grids and channel lists must use virtualization/lazy loading; target first meaningful paint under 2s on a mid-tier mobile connection.
Availability	Target 99%+ uptime for the web app and API during the free MVP period, acknowledging free-tier hosting limits.
Scalability	Architecture must comfortably support the confirmed MVP target of 500–5,000 registered users without redesign; horizontal scale path documented in Document 3.
Security	PINs stored as salted hashes, never plaintext; XC credentials encrypted at rest; role-based access control strictly enforced (End User / Reseller / Super Admin data boundaries per FR-6.1).
Usability	Dark-first UI with glassmorphism accents, 16–20px rounded cards, skeleton loaders, and shimmer placeholders during playlist/content loading.
Responsiveness	Fully responsive across mobile, tablet, and desktop browser viewports; collapsible sidebar on desktop, adapted navigation on mobile.
Offline Behavior	Locally cached Device ID, last-known playlist metadata, and app shell available via service worker even when network is briefly unavailable; live streaming itself requires connectivity.
Compatibility	Modern evergreen browsers (Chrome, Edge, Safari, Firefox) on Android, iOS, Windows, macOS, Linux.
Data Integrity	Duplicate detection on playlist save; referential integrity between Users, Playlists, Favorites, and History (formalized in Document 4).

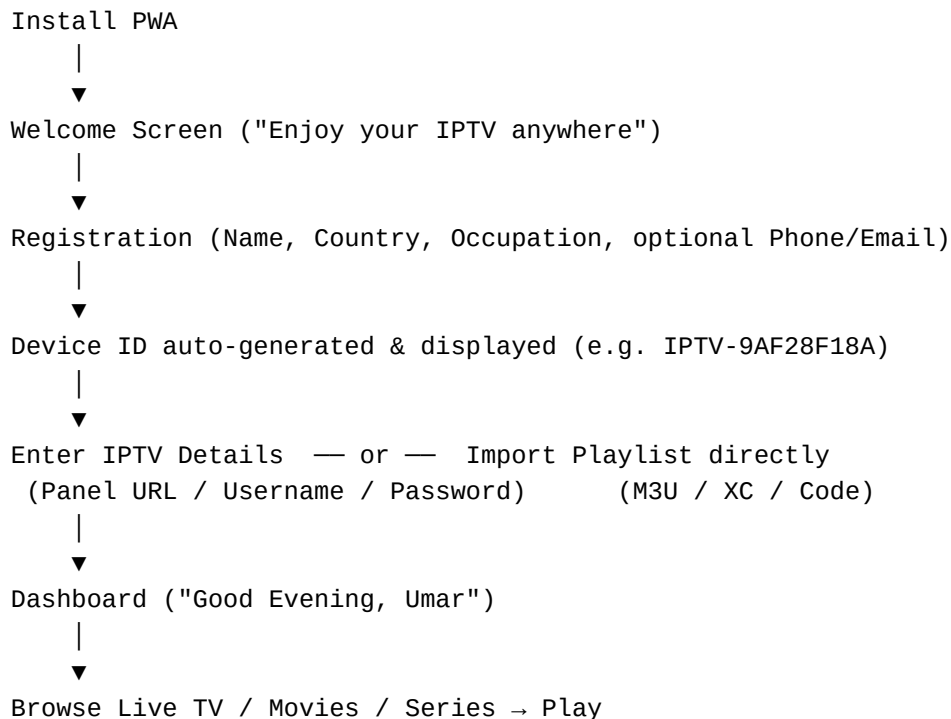
10 9. Product Features (Summary Matrix)

Feature	End User	Reseller	Super Admin
Registration & onboarding	Yes	Yes (own account)	Yes (own account)
Add/Edit/Delete own playlists (M3U/XC/Code)	Yes	—	—
PIN-protect a playlist	Yes	—	—

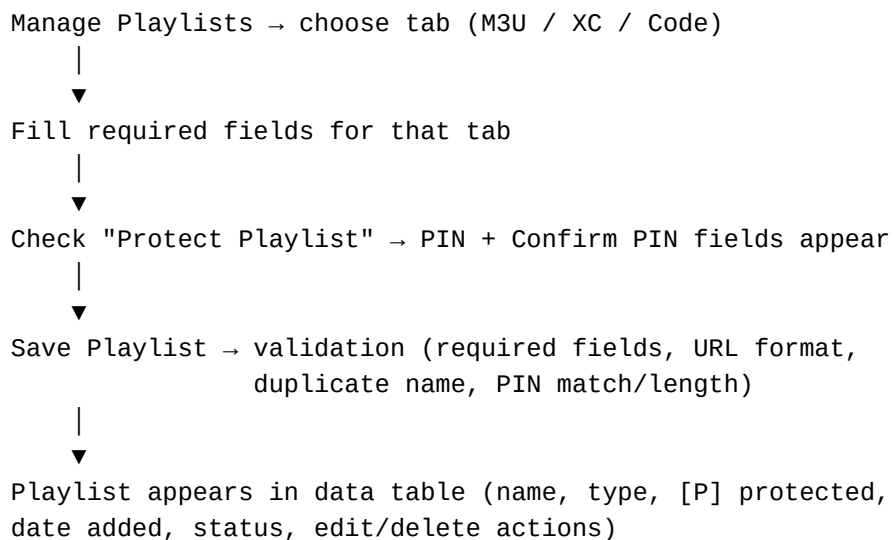
Feature	End User	Reseller	Super Admin
Browse & play Live TV/Movies/Series	Yes	—	—
Favorites / Continue Watching	Yes	—	—
Global search	Yes	—	—
Manage own sub-users' playlists	—	Yes	Yes (all users)
View sub-user status/devices	—	Yes (own only)	Yes (all)
Platform analytics	—	—	Yes
Disable any account	—	—	Yes
Set future pricing/plans	—	—	Yes (post-MVP)

11 10. User Journeys

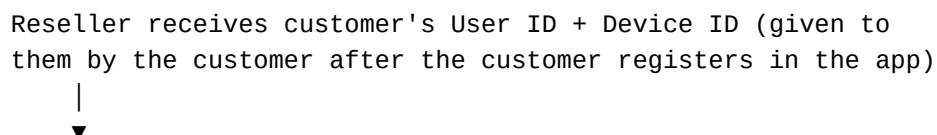
11.1 10.1 End User — First Install to First Playback



11.2 10.2 End User — Adding a Protected Playlist



11.3 10.3 Reseller — Onboarding a Customer



Reseller logs into Reseller Portal → finds/links that User ID

|



Reseller assigns a playlist (M3U / XC / Code) to that user

|



End User's app reflects the newly assigned playlist

|



If customer stops paying (handled outside BX Player during free period) → Reseller revokes/disables the playlist

11.4 10.4 Super Admin — Platform Oversight

Super Admin logs into Admin Portal

|



Dashboard: Online Users / Installed PWAs / Active & Expired Panels

|



Drill into any Reseller → see their End Users

or

Drill into any End User directly (cross-reseller visibility)

|



Assign / Revoke / Disable playlists or accounts

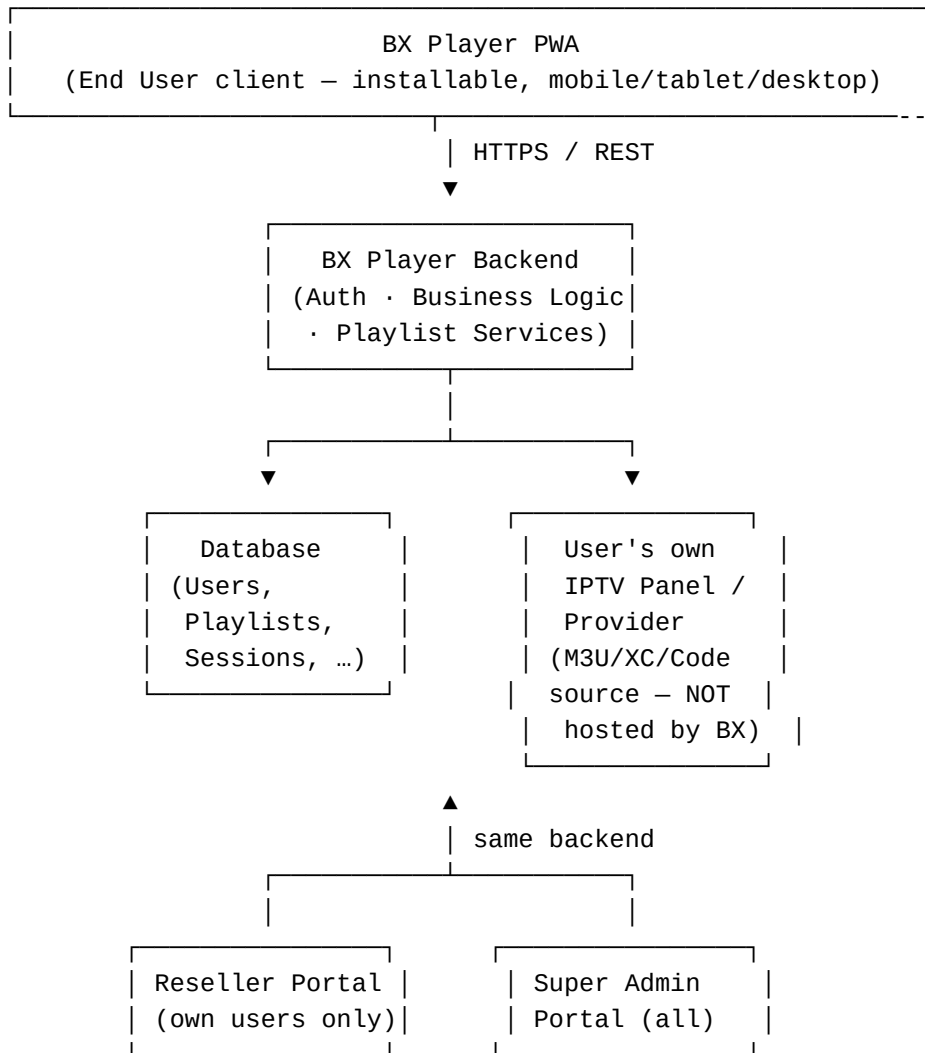
|



Review Analytics: DAU, Watch Time, Popular Channels, Buffer Errors, Device Types

12 11. System Workflow (High-Level)

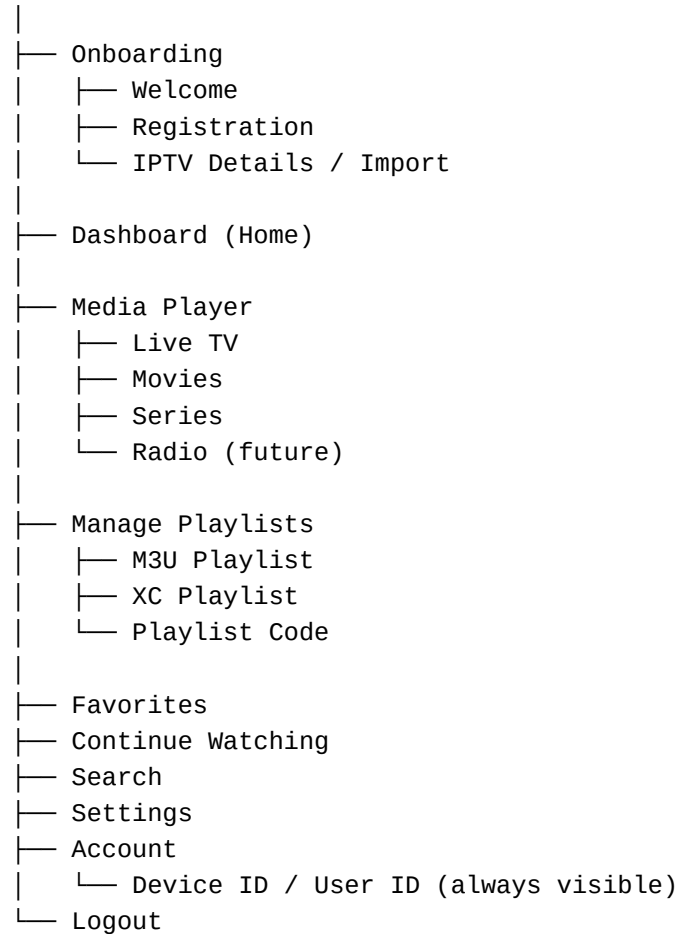
This is a conceptual workflow only — technical architecture, hosting topology, and deployment decisions are formally specified in Document 3 (System Architecture). It is included here to confirm shared understanding before design begins.



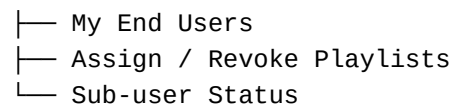
Key confirmed constraint: **BX Player never hosts or proxies the actual video content.** Playback streams directly from the user's own panel/provider; the platform's backend only manages accounts, credentials/metadata, and app-level state.

13 12. Information Architecture

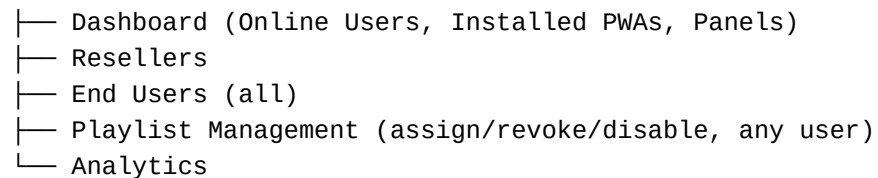
BX Player (PWA)



Reseller Portal



Super Admin Portal



14 13. Assumptions & Constraints

#	Assumption / Constraint
A1	Users already possess a working IPTV panel/subscription from a third party; BX Player does not source or resell content itself.
A2	Initial platform target is Web PWA only ; no native Android/TV app store submission in this phase (confirmed decision, revisitable later).
A3	Platform is free for ~6–7 months post-launch; no billing integration exists in the MVP, though the data model reserves fields for future plans.
A4	Reseller role is scoped strictly to managing their own onboarded End Users — no pricing/plan authority (confirmed decision).
A5	MVP infrastructure targets free/low-cost tiers (e.g., Vercel + Supabase/Neon) sized for the confirmed 500–5,000 user scale.
A6	XMLTV EPG ingestion is optional per playlist and depends on the URL the user supplies; BX Player does not generate EPG data itself.

15 14. Success Metrics (MVP Period)

Metric	Target Signal
Registered devices	Growing toward 500–5,000 within the free period
Reseller adoption	Multiple active resellers each managing a real sub-user base
Playback reliability	Low reported buffering/failure rate relative to source panel quality
Retention	Users returning via Continue Watching / repeat sessions
Admin usability	Super Admin able to locate and act on any user/reseller without needing direct database access

16 15. Glossary

Term	Definition
PWA	Progressive Web App — installable web application using a manifest and service worker.
M3U / M3U8	Playlist file formats listing IPTV stream URLs.
XC / Xstream Codes	A common IPTV panel API standard authenticated via Host, Username, and Password.

Term	Definition
Playlist Code	A provider-issued short code that resolves to a playlist server-side, without exposing a raw URL.
EPG / XMLTV	Electronic Program Guide data, typically supplied as an XMLTV-format URL.
Device ID	A unique identifier auto-generated per installation, used to identify the client device.
User ID	An identifier issued to/by the user, given to a Reseller/Provider to link a panel to the account.
Reseller	A platform role that onboards and manages a subset of End Users.
Super Admin	The platform-wide administrative role with full visibility and control.

17 16. Approval & Next Steps

This PRD defines **Phase 1** only. Per the confirmed development strategy, the next step is **Document 2 — UI/UX Design Specification**, which will detail every screen (wireframe-level), component, interaction, animation, and responsive behavior described functionally above — still using mock data, with no backend or database assumptions.

Before Document 2 begins, please confirm or amend:

1. Anything in Sections 4 (Scope) or 7 (Functional Requirements) that doesn't match your intent.
2. Whether the Reseller/Super Admin boundary in Section 7.6–7.7 is exactly right (especially FR-6.4).
3. Any additional MVP feature you want added to or removed from Section 4.1/4.2.

Completed: Document 1 — Product Requirements Document (PRD) **Next:** Document 2 — UI/UX Design Specification